Set Up an FTPS Server in Linux

vsftpd is a very popular package for FTP (File Transfer Protocol) but poses a security threat because it transfers vital data like usernames, passwords, etc, in plain text. To eliminate this risk, FTP offers encryption with the help of the SSL and TLS protocols. This article explains how to set up an FTPS server in Linux.

FTP is a standardised network protocol and probably the quickest as well as easiest option available when a large chunk of data is to be transferred from one host to another, over a TCP-based network. FTP defines a client-server architecture that uses two separate well-known ports for data (Port No 20, used for data transfer) and control (Port No 21, used for authentication) connections, in order to establish connectivity between the server and the client.

When it comes to the Linux operating system, the most popular package used to set up an FTP server is *vsftpd* or '*very secure FTP daemon*'. It offers very basic features such as anonymous enabling/disabling, local enabling/disabling and *chroot* jail for the users. But, when looked at from the security perspective, vsftpd has very few features to offer. Whenever the file transfer is initiated, all the data, including user credentials and passwords, get transferred in an unencrypted format, as plain text, which is considered to be very risky in any public network.

As a security measure, we have two options that offer secure file transfer capabilities, which are SFTP and FTPS. SFTP uses an SSH connection to run file transfers over a secure channel, while FTPS uses cryptographic protocols such as SSL (Secure Socket Layer) and TLS (Transport Layer Security). This article elaborates on the SFTP protocol in order to set up a secure FTP server using SSL certificates.

Installation of the required packages

To install openssl and vsftpd in Debian-based systems, you can run:

```
sudo apt-get install vsftpd
sudo apt-get install openssl
```

In Red Hat Linux-based systems, you can run:

```
yum install vsftpd
yum install openssl
```

Generating the SSL certificate and RSA key file

This step involves creating a SSL certificate file (*rsa_cert_file)* and RSA key file (*rsa_private_key_file*) that will be used by vsftpd for data encryption purposes. It is very important to set the paths of both these files, as those must be mentioned in the vsftpd configuration file (Red Hat -*/etc/vsftpd/vsftpd.conf* and Debian - */etc/vsftpd.conf)* in the *rsa_cert_file* and *rsa_private_key_file* variables. By default (in RHEL), the *rsa_cert_file* will point to */usr/share/ssl/certs/vsftpd.pem.*

For our convenience, let's put the certificate and the key in the same file, and store that file as */etc/vsftpd/vsftpd.pem.*

```
openssl req -x509 -nodes -days 365 -newkey rsa:1024 -keyout \
/etc/vsftpd/vsftpd.pem -out /etc/vsftpd/vsftpd.pem
```